

Analyzing Solar Imagery and Identifying Active Regions using Aditya-L1 data

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program Space Science Technology

**Astronomy & Astrophysics: Heliophysics with Aditya
L1 Data
By**

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Analyzing Solar Imagery and Identifying Active Regions using Aditya-L1 data



INDIA SPACE ACADEMY SUMMER TRAINING PROGRAM 2026

PROJECT REPORT

Analyzing Solar Imagery and Identifying Active Regions Using Aditya L1 Data

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ASTRONOMY & ASTROPHYSICS: HELIOPHYSICS WITH ADITYA-L1 DATA

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I also acknowledge the use of Aditya-L1 mission datasets and publicly available scientific resources that enabled the completion of this works.

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CERTIFICATE

This is to certify that the project entitled as”**Analyzing Solar Imagery and Identifying Active Regions Using Aditya-L1 Data**” has been carried out by **Shikshita Naryal** under the Summer Training Program 2026 conducted by
India Space Academy.

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ABSTRACT

For ages Sun has sustained all the living lives and so does has significantly impact the space environment, it's impact is directly seen near the earth. The presence of high magnetic field zones is essential for understanding solar flares, UV Brightnings, Coronal Mass Ejections.

Thee methodology involves several image processing techniques like: data acquisition in FITS format, normalization, gaussian smoothing andregion labeling. These methods evaluate the performance and validate against the established solar records.

The research highlights the projects's success in providing precise solar measurements.

The result reinforces how Aditya-L1 data contributes significantly to modern heliophysics research and thus advancing solar physics.

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Acronyms

ISRO – India Space Research Organization

PRADHAN Portal – Program for Analysis of Data and Archival Network

UV – Ultraviolet

FITS – Flexible Image Transport System

L1 – Lagrange Point 1

CME – Coronal Mass Ejection

VELC– Visible Emission Line Coronagraph

SUIT – Solar Ultraviolet Imaging Telescope

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Introduction

Since we know that sun is the brightest, nearest and central star of our solar system. It provides almost all of our energy and it's activity strongly affects the environment around the space and earth. So understanding the activities taking place in and around sun is very crucial as it influences the satellites, communication systems, space weather conditions, etc.

When the sun's magnetic field interact with highly mobile charged particles in it gives rise to a variety of observable events. These events are collectively known as solar activity.

Solar Flares are the sudden eruptive events involving energy in the range 10^{22} Joules to 10^{25} Joules. The most places of occurrence of solar flares are the regions of closely placed sunspots (dark spots on the solar disk) and they are possibly caused due to the release of the stored magnetic energy.

The solar atmosphere is divided into: Photosphere, Chromosphere, Transition Region and Corona. Among these layers there are localized regions of concentrated magnetic flux known as Active Regions and this is the region where Solar Flares, Coronal Mass Ejections, Enhanced Ultraviolet or X-Ray takes place. To detect and investigate on these high resolution space based observations are required that are free from the absorption and distortion of earth's atmosphere.

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Overview: Aditya-L1 Mission

Aditya-L1 is India's 1st dedicated space based observatory for studying the sun launched by India Space Research Organization (ISRO) on September 2, 2023. The mission is actively generating the critical scientific data as from June, 2026.

This Mission Carries Seven Indigenous Payloads to study the Sun's outer layer.

Remote Sensing Payloads:

- VELC (Visible Emission Line Coronagraph): It studies the solar corona and the dynamics of Coronal Mass Ejection.
- SUI (Solar Ultraviolet Imaging Telescope): It is used to study the transition region and lower corona. It images the solar atmosphere in and near the UV region.
- SoLEXS & HELIOS: It has soft and hard X-ray spectrometer to observe solar flare activities.

In-situ Payloads:

- ASPEX & PAPA: It analyze the solar wind components and it includes ions and energetic particles.
- MAG: It is a high resolution digital magnetometer to measure the interplanetary magnetic field at L1.

ISRO has released over 27 terabytes of open access data to the global scientific community.

In 2026, the mission has entered a "First AO Cycle"(April-June), specifically inviting Indian Researchers to propose new observations using the VELC and SUI payloads.

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Solar Active Regions

Active Regions refer to the localized sectors on the solar surface characterized by highly intense, complex and dynamic magnetic structures. These magnetic complexes are generally present as bipolar or multipolar configurations.

They originate deep within the solar interior and rise outward due to magnetic buoyancy eventually penetrating the photosphere to extend into the upper regions of the solar atmosphere.

Within the photosphere, these areas manifest visually as dark sunspot clusters. This darkening occurs because the dense, concentrated magnetic field lines which actively suppress local convective heat transport.

Conversely when we observe higher up in the chromosphere and corona then these active regions are marked by the structural features known as active loops.

Driven by localized magnetic reconnection and intense plasma heating, these structures radiate vividly across ultraviolet wavelengths.

So Solar Active Regions serve as the primary driver behind the Sun's most volatile phenomena acting as a birthplace for high energy events like Solar Flares, CME's.

The Aditya L1 spacecraft secures an uninterrupted, eclipse free vantage point. This positioning allows its remote sensing payloads.

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Literature Review

Since sun is a dynamic star whose activities continuously affects the surrounding space environment using different phenomenons either by sunspots, active regions, solar flares and CMEs. So studing these features is important to study there effect on satellite operations as a result solar image analysis has become a major area of research in modern astrophysics.

Aditya L1 is first dedicated solar mission developed by ISRO that is positioned near Sun-Earth Lagrange point 1 i.e. L1. SUT captures UV images of sun , proving valuable information about solar structures and activity.

The availability of Aditya L1 data has opened the new opportunities for studying the sun using the different image processing techniques. Research has shown that active regions store substantial magnetic energy. So identifying and tracking the active regions is a critical step in understanding solar behaviour.

I have first downloaded the FITS files from ISRO PRADHAN Portal and then wrote the code which preprocesses the images (i.e. FITS files) and normalized them.

Then used the tools like Gaussian filtering, threshold based segmentation and region labeling technique for solar activity region detection. My present work builds upon using these tools to approach the results for indentification and visualization of active regions associated with enhanced solar activity.

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METHODOLOGY

PART-1 (MARKS-45)

OVERVIEW

Part 1 has primary objective to identify Solar Active Region from UV solar imagery using image processing techniques. To achieve the objective the acquired solar images from ISRO PRADHAN Portal are preprocessed segmented and then analyzed using automated detection method.

The complete workflow consisted of six major stages:

1. Data Acquisition
2. Data Loading
3. Image Preprocessing
4. Active Region Detection
5. Region Labelling and feature Extraction
6. Visualization and validation

Since the data mentioned in Project work File (**Data should be of Dec 31 2023, Flare strength is X5.**) was not available in ISRO PRADHAN portal so data of 22 February, 2024 was picked as it was having flare strength X6.3.

Dataset File Used From SUIT



Input Data

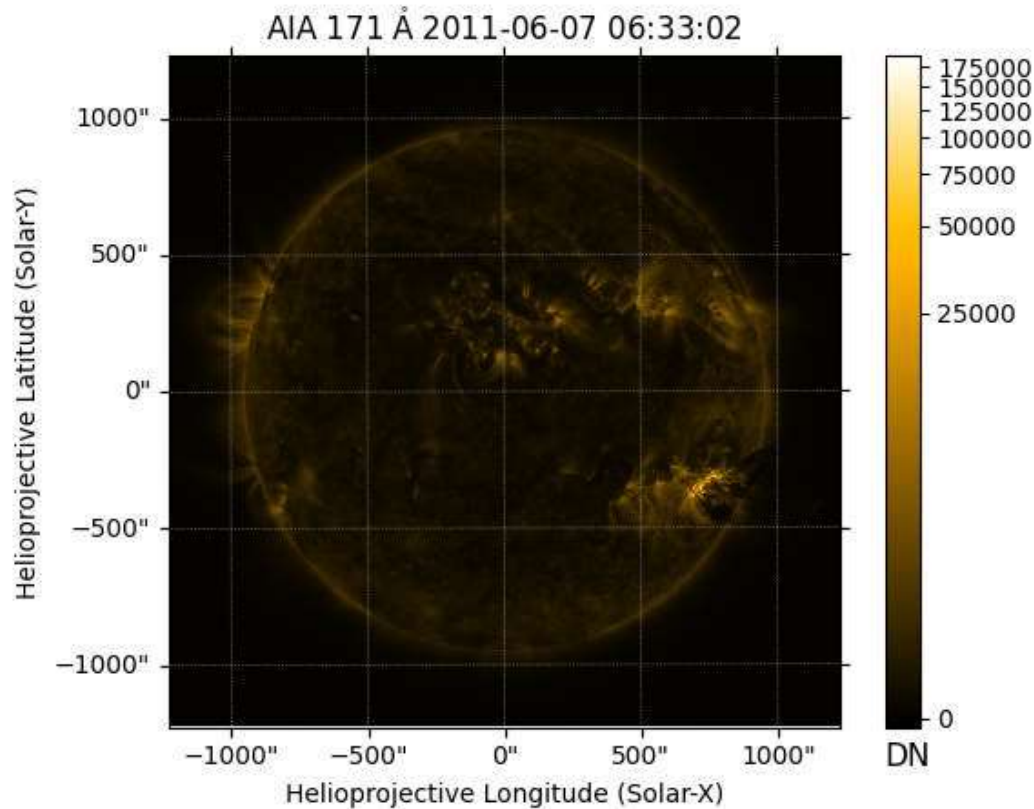
To analyze the FITS format solar image was loaded to the code using SunPy library. This library ensures the successful execution of the analysis pipeline.

The input image contains the intensity values recorded by SUIT across the solar disk and each pixel represents the brightness of a specific solar location. The image serves as the base dataset from which the active regions are extracted and analyzed.

Output:

A solar intensity map is ready for further scientific processing.

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Step 1: Data Acquisition

My project is designed for observing the Solar activity period of 22 February 2024 of strength X6.3

In this step we obtain solar observational data which is suitable for active region identification.

Result: A solar image was successfully loaded and prepared for further processing.

Step 2: Image Preprocessing

The raw solar images which contains the varied colour intensity and have Vried brightness scales is processed.

We normalize the raw images so that the visibility of solar structure is improved, making the thresholding more reliable.

It also reduces the dependency on instrument specific brightness scales.

Output:

The figure of solar disk after normalization is being displayed.

Bright regions appear more prominent and it becomes easier to detect them during segmentation.

The normalized image clearly highlights the bright solar structure while strill maintaining the overall solar disk information.

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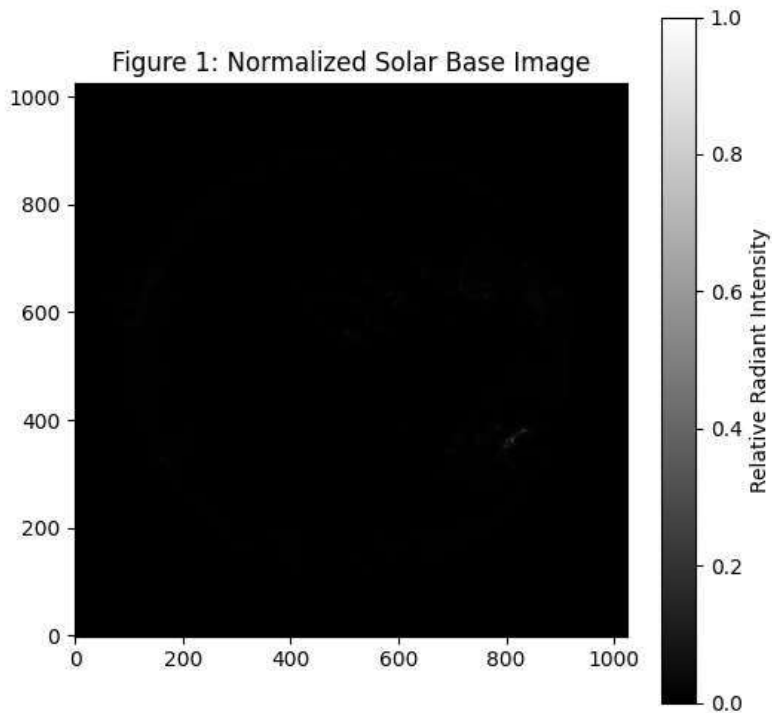


Fig.1 Normalized Solar Base Image

Step 3: Detecting Active Regions

We need a technique to distinguish the active regions from less active background areas.

Otsu's algorithm is being used in my code which automatically calculates an optimal threshold value based on image intensity distribution.

The threshold was slightly reduced by multiplying it by 0.9 to improve the sensitivity and to allow the detection of weaker active regions.

Code snippet:

```
threshold = filters.threshold_otsu(data_norm)
# Adjusting the global baseline of threshold to safely isolate faint magnetic
complexes
adjusted_threshold = threshold * 0.9
binary_mask = data_norm > adjusted_threshold
```

Pixels with intensity values greater than the calculated threshold value were classified as active region.

Result:

Image was generated where :

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White pixels represents the potential active regions

Black pixels represents background solar areas.

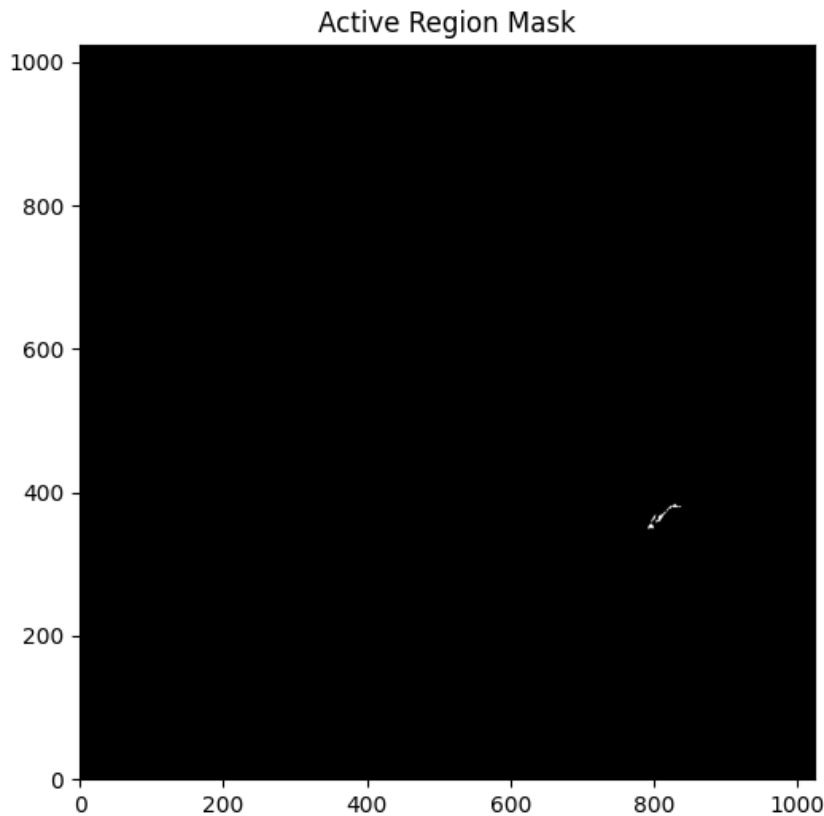


Figure: Active Region Mask

Step 4: Morphological Noise Removal

After thresholding the small isolated bright pixels may still exist due to the noise. So to eliminate the tiny disconnected regions that don't represent the actual solar activity.

So it removes the false detections, improves segmentation quality and produces clearer region boundaries.

Output:

The image obtained contains only significant active region structure while most of the noise is being removed.

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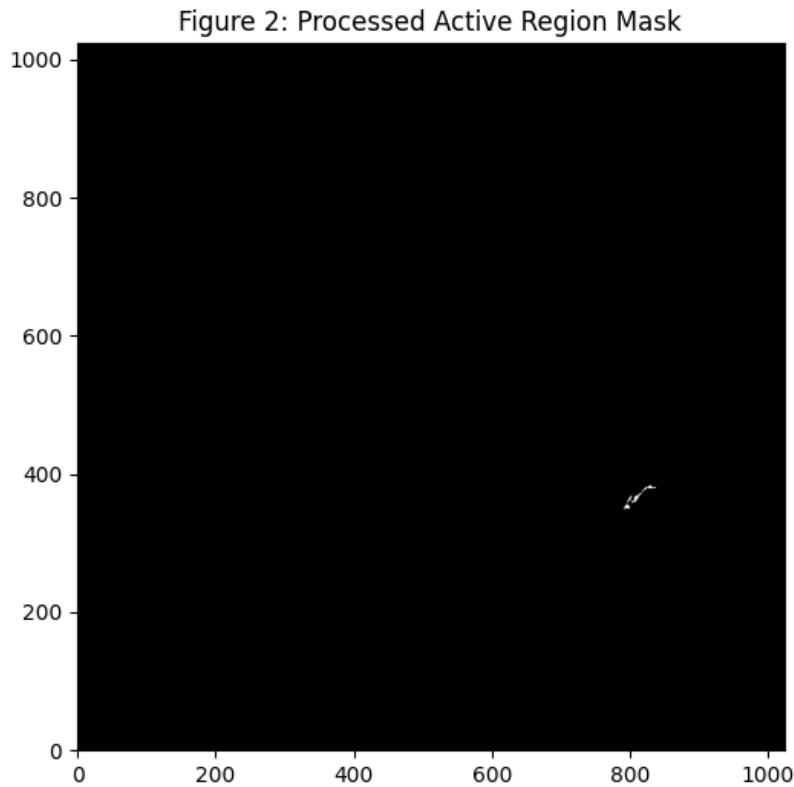


Fig.3 Processed Active Region mask

Step 5: Connecting Component Labeling

Once the active region mask was generated the connected regions were identified automatically. This is used to separate the individual active regions from another. This algorithm scans the binary image and assigns a unique label to each connected structure. The each region is treated as an independent solar active region.

So each deected region gives

1. Area
2. Boundary coordinates
3. Sape information
4. Position on the solar disk

So the algorith outputs the total number of detected active regions

Step 6: Active Region Visualization

To maake the results easier to interpret we drow the rectangles to bounday of each detected region.

We highlight vthe location and region of each active region.

Output:

The red coloured rectangles indicates the aras where the code detected the solar activity. The larger rectangles

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corresponds to larger active regions with great spatial coverage.

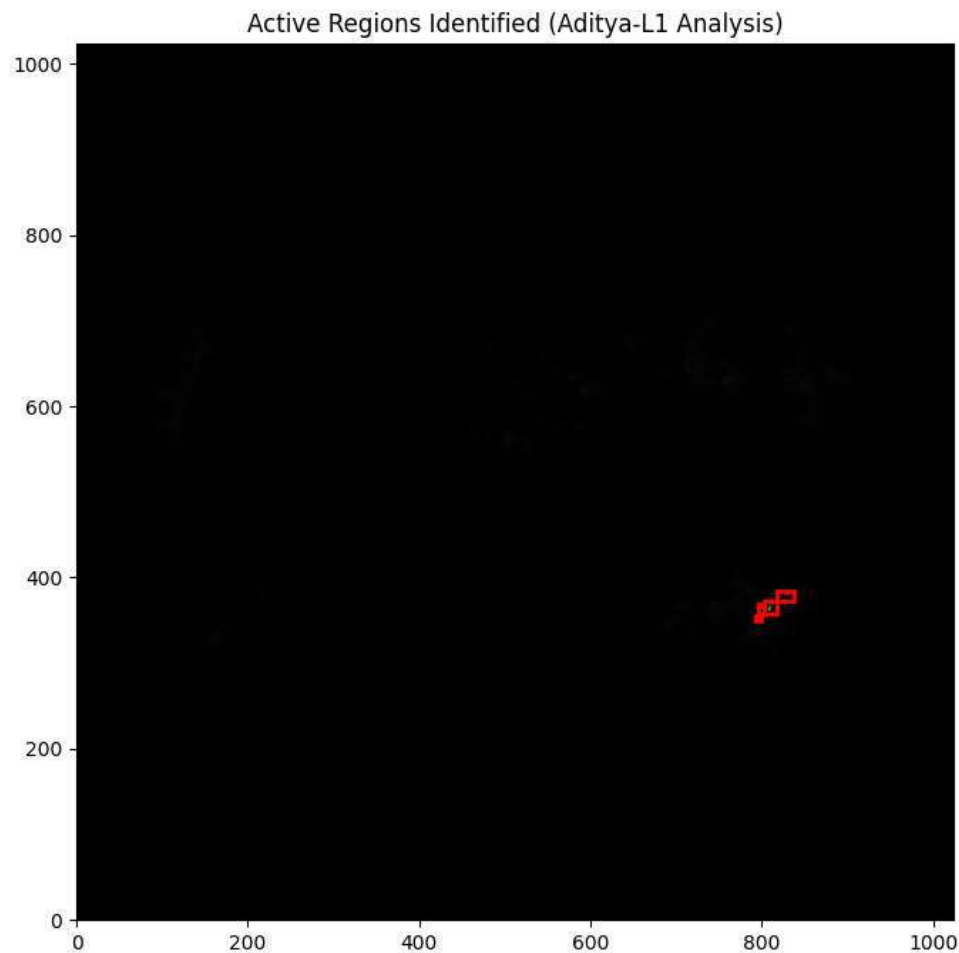


Figure4: Active regions indentifies using boundary boxes

Step 7: Validation Method

Now the active regions detected corresponds to the actual solar active regions.

-> Analysis Metrics: Detected 4 independent active regions.

Detected 4 active regions.

Conclusion:

For part 1 of the project report this automated methodology provides an automated and highly efficient workflow for identifying solar active regions. Mix of normalization, thresholding and filtering the raw FITS files to clean up the data from the images and isolate the active regions.

The resulting figures in output thus shows the processed images so that next work part of the project can smoothly handle the next task. We analyzed the portion where solar flare activity took place.

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So this part 1 is the foundation for the next part 2 by providing the necessary data, segmented images and feature extractions for tracking of these regions over time, studying temporal evolution, intensity magnetic flux correlation, comparative segmentation method would be observed.

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Collab Link for my Python Code:

<https://colab.research.google.com/drive/1a60Ynns5Bbf7fBJwNT-YNBi4wjbjFrxm?usp=sharing>

Downloaded Python File :

[AdityaL1_ShikshitaNaryal_Project_Part1,2\(A,B,C\),3.ipynb](#)



AdityaL1_ShikshitaNar
yal_Project_Part1,2(A,B,C),3.ipynb

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